

Dataset	ε	Nodes	$C_{\max}$	GPU FP32			CPU FP64		
				Time (s)	Δ_{GPU}	Pass	Time (s)	Δ_{CPU}	Pass
GSE24080	Exact degree	27,338	0	1.172 ± 0.035	0	—	69.277 ± 2.833	0	—
	10^{-1}	44–53	7.69×10^{-2}	0.090 ± 0.020	3.40×10^{-4}	5/5	0.156 ± 0.013	2.36×10^{-9}	5/5
	10^{-2}	45–54	7.09×10^{-3}	0.085 ± 0.007	3.71×10^{-4}	5/5	0.164 ± 0.012	2.36×10^{-9}	5/5
	10^{-3}	46–54	8.82×10^{-4}	0.086 ± 0.006	1.31×10^{-3}	4/5	0.162 ± 0.010	2.36×10^{-9}	5/5
TCGA LGG	Exact degree	198,033	0	66.377 ± 1.196	0	—	3656.562 ± 12.642	0	—
	10^{-1}	37–49	7.01×10^{-2}	0.320 ± 0.028	4.47×10^{-3}	5/5	0.992 ± 0.076	2.44×10^{-7}	5/5
	10^{-2}	38–50	6.51×10^{-3}	0.312 ± 0.006	3.80×10^{-3}	5/5	1.009 ± 0.097	2.44×10^{-7}	5/5
	10^{-3}	39–51	5.80×10^{-4}	0.310 ± 0.011	5.95×10^{-3}	2/5	1.035 ± 0.100	2.44×10^{-7}	5/5

Table 1: Cox explanations with 30 fixed training-background samples and five distinct held-out patients per cohort on an Apple M4 Pro. Times are mean $\pm$ sample standard deviation across patients, with one synchronized measurement per patient and method. $C_{\max}$ is the largest analytical absolute quadrature-error bound across patients. Δ_{GPU} is the largest absolute feature-wise difference from the matching GPU FP32 degree-exact output; Δ_{CPU} uses the matching CPU FP64 degree-exact output. Both maxima range over all five patients and features. Pass counts patients whose difference from their same-backend reference is at most ε . Exact-row differences are zero by construction, not evidence of zero floating-point error. CPU degree-exact measurements are new; GPU results and CPU approximate measurements reuse the matched saved runs. Warm-up and quadrature-rule construction are excluded; approximate API times include automatic node selection. Previously measured exact-rule construction required 10.97 s for GSE24080 and 561.92 s for TCGA; approximate-rule construction required 0.386–0.563 ms. CPU exact timings measure active checkpointed integration with the same factor/block order as the public API, excluding checkpoint I/O and preliminary kernel-shape pilot calls. The certificate excludes rounding, and agreement with a rounded reference is not an absolute numerical guarantee.

1.1 High-dimensional experiments with QuadraSHAP

To assess whether Shapley attribution remains practical when the number of features greatly exceeds the number of patients, we consider GSE24080 multiple-myeloma gene expression and TCGA lower-grade glioma DNA methylation [1, 2]. Genome-wide assays yield 54,675 expression probe sets and 396,065 methylation sites, respectively, whereas the training cohorts contain only 339 and 383 patients. We fit ridge-penalized Cox models and explain their relative hazard, whose feature-wise product structure makes QuadraSHAP directly applicable. Each explanation uses the same 30 training-background patients; results summarize one timed call for each of five distinct held-out patients. We compare degree-exact quadrature with automatically selected node counts for $\varepsilon \in \{10^{-1}, 10^{-2}, 10^{-3}\}$ on an Apple M4 Pro, using GPU FP32 and CPU FP64 arithmetic.

Table 1 shows that degree-exact GPU computation remains manageable: 27,338 and 198,033 nodes require 1.17 s and 66.38 s per patient, respectively. The certified approximation reduces these latencies to 85–90 ms and 310–320 ms, supporting interactive, on-demand explanations after warm-up and quadrature-rule setup. CPU FP64 provides a measured alternative when higher numerical precision is required, with approximate latencies of 156–164 ms and 992–1035 ms, respectively.

For a matched numerical comparison, each approximation is compared with the degree-exact output from its own device and precision. All selected rules satisfy their analytical quadrature bounds, but FP32 arithmetic limits agreement at the tightest tolerance: at $\varepsilon = 10^{-3}$, 4/5 GSE24080 and 2/5 TCGA GPU cases meet that threshold. CPU FP64 meets its same-backend threshold in 30/30 cases. The guarantee concerns quadrature for the fixed fitted model and empirical background; it excludes rounding and population-background approximation. Agreement with degree-exact floating-point output is therefore an empirical consistency check, not an absolute error proof. Full settings, timing provenance, and error definitions are given in Appendix A.1.

A Additional experimental details

A.1 Additional details for high-dimensional Cox experiments

To assess whether Shapley attribution remains practical when the number of features greatly exceeds the number of patients, we consider two genomic survival datasets: GSE24080 multiple-myeloma gene expression [1] and TCGA lower-grade glioma (LGG) DNA methylation [2]. Genome-wide assays measure many molecular variables per patient, whereas the cohorts with matched survival follow-up contain only hundreds of patients. After eligibility filtering, GSE24080 contains 553 patients and 54,675 expression probe sets, and TCGA LGG contains 511 patients and 396,065 methylation sites. These are probe- or site-level features, retained without gene aggregation. We use the supplied 339/214 training/validation split for GSE24080 and a 383/128 split stratified by event status for TCGA LGG (seed 42), giving an extreme small- n , large- d setting.

We model overall survival using a Cox proportional-hazards model, which incorporates right-censored observations: patients whose death has not been observed by the end of follow-up. Its relative hazard factors across features, making QuadraSHAP applicable directly:

$$h(t \mid x) = h_0(t)f(x), \quad f(x) = \exp(\beta^\top x) = \prod_{j=1}^d \exp(\beta_j x_j).$$

To stabilize estimation when $d \gg n$, we shrink the Cox coefficients towards zero using a ridge penalty, with the implementation’s penalty parameter fixed a priori at $\alpha = 0.01d$. We use the Breslow approximation for patients with equal recorded event times. Missing values are imputed using training medians, and centering and scaling use training statistics. All retained features have nonzero fitted coefficients.

We explain the relative hazard using 30 training-background patients, sampled without replacement with seed 42 and held fixed across methods. Specifically, absent features are replaced by the corresponding entries of each background row and predictions are averaged:

$$v_x(S) = \frac{1}{30} \sum_{b=1}^{30} f(x_S, z_{\bar{S}}^{(b)}).$$

The resulting game is a weighted sum of product games. We explain the first five held-out patients in each saved split with the degree-exact rule $m_q = \lceil d/2 \rceil$ and with patient-specific node counts selected to certify absolute quadrature tolerances $\varepsilon \in \{10^{-1}, 10^{-2}, 10^{-3}\}$.

Matched degree-exact references. To assess approximation accuracy without changing the numerical backend, we use a separate degree-exact reference for each device/precision setting. Write $\hat{\phi}_{m_p, p}^a$ for patient p ’s computed attribution vector using m_p nodes, with $a = \text{GPU}$ denoting GPU FP32 and $a = \text{CPU}$ denoting CPU FP64. Let $m_\star = \lceil d/2 \rceil$. The table reports

$$\Delta_a = \max_{p=1, \dots, 5} \|\hat{\phi}_{m_p, p}^a - \hat{\phi}_{m_\star, p}^a\|_\infty.$$

Thus GPU approximation is compared only with GPU FP32 degree-exact output, and CPU approximation only with CPU FP64 degree-exact output. These are full degree-exact rules with 27,338 and 198,033 nodes, respectively; the CPU references are newly measured rather than replaced with smaller-node FP64 proxies. A patient passes if its maximum absolute feature-wise discrepancy is at most its requested tolerance. Exact-row discrepancies are zero by self-comparison and do not assert zero rounding error.

For the exact real-arithmetic Shapley vector ϕ_p and its quadrature approximation $\phi_{m_p, p}$, the analytical node selector instead guarantees

$$\|\phi_{m_p, p} - \phi_p\|_\infty \leq C_p \leq \varepsilon, \quad C_{\max} = \max_{p=1, \dots, 5} C_p.$$

This bound can be conservative: the actual quadrature error need not be close to either C_p or ε . However, both computed vectors in Δ_a also contain floating-point error. Consequently, a discrepancy above ε need not contradict the quadrature certificate, while a discrepancy below ε does not independently establish an absolute error guarantee for either computed vector.

To avoid choosing unnecessarily many nodes, the implementation searches integer node counts $m = 1, 2, \dots$ and returns the first whose analytical bound meets the requested tolerance; the degree-exact threshold is the fallback cap. Evaluating the bound for each candidate requires optimizing an auxiliary scalar that controls the bound. This inner optimization solves its scalar stationarity equation by bisection, repeatedly halving an interval containing the root. The separate `closed_form_budget` calculation is recorded as a diagnostic and does not select the node count used in these experiments. Timed automatic node selection includes both the patient/background factor-summary pass and this integer search.

Implementation and timing. GPU explanations use JAX-Metal FP32; CPU explanations use JAX with FP64 enabled. Both evaluate the same parallel product-game formula, use an 8 GiB memory-planning budget and one background row per block, and accumulate returned blocks on the host in FP64. The memory setting is a planning budget, not a measurement of actual allocation. Each table entry summarizes five distinct held-out patients, with one timed explanation per patient and method. We report the arithmetic mean and sample standard deviation across those patients. This variation is not repeated-run uncertainty and does not establish statistical significance.

GPU exact and approximate measurements are reused from the matched saved experiments, and CPU approximate timings reuse the warmed CPU FP64 sweep. The full degree-exact CPU runs are new. Model coefficients, preprocessing, background and held-out rows, numerical source hashes, and package versions are verified to match. Timings exclude rule construction and recorded warm-up. CPU degree-exact computation uses a checkpointed integration loop with the same factors, node order, and background/block accumulation order as the public API; a small synthetic comparison was bitwise identical. Before each dataset, one actual node block with the full exact-core shape is evaluated for warm-up, followed by three pilot core calls. These calls are excluded; the warm-up is not a full degree-exact explanation. The reported CPU exact time sums active integration, including factor preparation, block planning, transfers, synchronized core evaluation, and host accumulation. Checkpoint I/O is measured separately and excluded; if resumed, inactive downtime and uncheckpointed work lost to interruption are also excluded. Thus this timing is not an undifferentiated process wall-clock measurement. Approximate API timings include automatic node selection as well as explanation, synchronization, transfers and host work. Measurements were recorded in separate runs. GPU FP32 versus CPU FP64 changes both device and precision, so their latency ratio is not a matched-precision GPU speedup.

The quadrature nodes and weights are cached across patients. Previously measured CPU construction of the exact GSE24080 and TCGA rules required 10.97 s and 561.92 s, respectively; the latest approximate rules required 0.386–0.563 ms (range of medians over five fresh constructions per rule). These setup costs are separate from explanation latency. Caching rules does not remove evaluations of the patient- and background-dependent integrands at those nodes.

Numerical interpretation and scope. All 30 selected approximate rules satisfy $C_p \leq \varepsilon$. GPU FP32 meets its same-backend discrepancy threshold in 26/30 cases, while CPU FP64 meets it in 30/30 cases. The largest CPU discrepancy across patients, features, and the three tolerances is 2.44×10^{-7} . The CPU FP64 results provide empirical evidence that the selected rules agree closely with full degree-exact quadrature at higher precision. They must be paired with the CPU runtime, not assigned to GPU FP32 timing. The numerical-reference convention in this table is distinct from earlier checks against a shared, independently implemented 64-node FP64 reference. Those checks and the separate CPU FP32 control are retained as diagnostic artifacts; they are not the definitions of either discrepancy reported here.

Finally, the explanations concern the fitted relative-hazard model and the fixed uniform distribution on the 30 selected background rows. The certificate does not quantify model-estimation uncertainty,

uncertainty from choosing a finite background sample, or discrepancy from a population background distribution. The survival datasets demonstrate computational feasibility in a small- n , large- d setting; these experiments do not establish clinical usefulness or causal interpretation of the attributions.

References

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